

Project 7: Improving DDPM. Does a Cosine Schedule Let a Diffusion Model Sample in Fewer Steps?

Group 4

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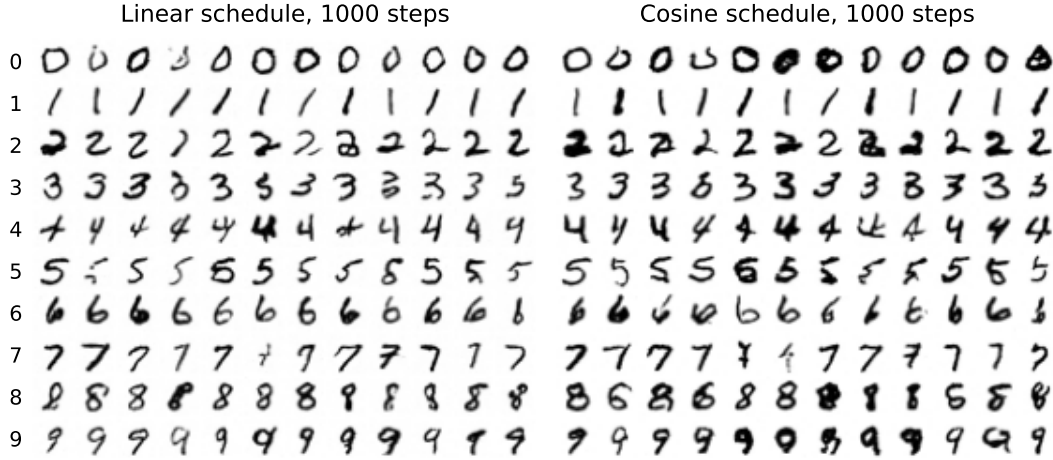


Figure 1: Twelve samples per digit from the linear-schedule (left) and cosine-schedule (right) models at seed 0 and 1000 steps; images in the same place start from the *same* noise. The classifier reads 0.9583 (left) and 0.9250 (right) as the intended digit.

1. Introduction

A denoising diffusion probabilistic model (DDPM; Ho et al., 2020) destroys an image in T small Gaussian steps and trains a network to predict the noise added, so that sampling can run the process backwards from pure noise. The *noise schedule* $\beta_1, \dots, \beta_T$ sets how fast the share of signal left, $\bar{\alpha}_t = \prod_{s \leq t} (1 - \beta_s)$, falls. Ho et al. raised β_t linearly from 10^{-4} to 0.02 over $T = 1000$ steps. Nichol and Dhariwal (2021) observed that the end of this process is “too noisy” to contribute much to sample quality, and proposed a cosine schedule that destroys information more slowly. Here 32.7% of the linear schedule’s steps leave less than 1% of the signal, against 6.5% for the cosine schedule, so a sampler that visits only K of the 1000 steps wastes fewer of them on near-pure noise. Hence the hypothesis: “A cosine schedule allows a denoising diffusion probabilistic model to generate high-quality samples using fewer time steps than a linear schedule.”

Nichol and Dhariwal found that a linear-schedule model “does not get much worse” when up to 20% of the reverse process is skipped. With the simple loss used here, their cosine schedule improved log-likelihood on CIFAR-10 (3.37 to 3.26 bits/dim) but not FID (2.90 to 3.05), and their sampling in far fewer steps came from *learning* the reverse variances; with fixed variances, as here, quality fell much faster as steps were removed. Song et al. (2021) showed that DDIM, a deterministic sampler for the same trained network, gives good samples 10 to 50 times faster. Lin et al. (2024) showed that common schedules and step choices leave signal at the last step, which harms sampling in few steps. None of them isolates what the schedule alone does to few-step sampling, which is the question tested here.

2. Methodology

Baseline. Tutorial 9.3’s conditional DDPM, unchanged: UNet_cond (221,569 parameters; its two poolings take $20 \rightarrow 10 \rightarrow 5$, so 20×20 images fit), with t/T and the class each embedded as 32 numbers added at every level. It predicts the added noise (mean squared error), trained with Adam (2×10^{-4}) in batches of 128 for 150 epochs on the 60,000 training images scaled to $[-1, 1]$, with $T = 1000$.

Cosine schedule. The only change: $\bar{\alpha}_t = f(t)/f(0)$

with $f(t) = \cos^2\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)$, $s = 0.008$, and $\beta_t = 1 - \bar{\alpha}_t/\bar{\alpha}_{t-1}$ clipped at 0.999 (Nichol and Dhariwal, eq. 17). Training is otherwise identical: for seeds 0, 1 and 2, both start from the same weights, batches, steps and noise.

Sampling in K steps. A K -step sampler visits K evenly spaced steps rounded to integers (Nichol and Dhariwal, §4) and jumps from t to the next visited t' with $\beta = 1 - \bar{\alpha}_t/\bar{\alpha}_{t'}$. Two samplers use it: the tutorial’s stochastic step (DDPM) and the deterministic DDIM step. Both form the clean-image estimate $\hat{x}_0$ from the predicted noise and clip it to $[-1, 1]$, as Ho et al.’s released sampler does; unclipped at $K = T$ the step equals the tutorial’s to rounding error. The clip matters for the cosine schedule, whose last β is 0.999: the tutorial’s form multiplies an error in the first noise estimate by 31.59 at $K = 1000$ and 3,492.67 at $K = 10$ (linear: 2.54).

3. Experiments

E1 (Figure 1): 12 samples per digit from each seed-0 model at $K = 1000$. **E2** (Table 1, Figure 2): each seed-0 model at $K = 10, 20, 50, 100, 250$ with both samplers and at 1000 with DDPM, 1,000 samples per set, drawn under the labels of a class-stratified test subset A from the same starting noise for both schedules. *Quality* is the Fréchet distance (FD; Heusel et al., 2017) to a disjoint real subset B with the same class mix, in the 128 features of a separate classifier (0.9914 test accuracy), as Salvado (2026, pp. 42, 87) proposes; real A against B gives the floor, 7.65. *Class consistency* is that classifier’s accuracy on the intended labels, and *variety* the within-class feature spread relative to real digits. **E3:** seeds 1 and 2 at every $K < 1000$. **A rule fixed in advance:** at a given K and sampler, a schedule is better if its FD is lower at seed 0 with a 95% paired bootstrap interval (1,000 resamples) excluding zero, and lower at seeds 1 and 2 too. **E4:** seconds per epoch, and the FD of seed 0’s snapshots at epochs 10, 25, 50, 100 and 150 (DDIM, $K = 50$). **E5:** each model’s noise-prediction error on test images at 20 fixed steps. **E6,** run after the verdict: the seed-0 DDPM runs with Ho et al.’s smaller posterior variance $\hat{\beta}$.

4. Discussion

Neither grid in Figure 1 shows mode collapse: every row reads as its digit, with varied slant, width and stroke

K	DDPM sampler				DDIM sampler			
	Linear	Cosine	Δ [95% CI]	Δ , seeds 1, 2	Linear	Cosine	Δ [95% CI]	Δ , seeds 1, 2
10	186.26	85.35	-100.91 [-111.11, -89.05]	-125.94, -120.97	49.58	106.91	+57.33 [+44.97, +69.68]	+3.86, -4.93
20	61.77	26.55	-35.22 [-41.54, -28.39]	-13.18, -29.46	33.89	40.78	+6.89 [-0.42, +14.88]	+16.94, -4.85
50	26.55	17.25	-9.30 [-13.84, -4.54]	+14.59, -2.90	28.48	27.31	-1.17 [-7.22, +5.27]	+25.26, -2.12
100	18.67	15.61	-3.06 [-7.57, +1.53]	+17.78, +2.57	27.28	25.13	-2.14 [-7.76, +3.97]	+26.75, -1.24
250	15.27	15.58	+0.31 [-3.64, +4.64]	+22.65, +1.28	26.80	24.09	-2.71 [-8.06, +3.21]	+27.35, -0.82
1000	17.12	13.50	-3.61 [-7.70, +0.74]	n/a	n/a	n/a	n/a	n/a

Table 1: Fréchet distance to real test digits (lower is better; real-data floor 7.65) at K sampling steps, seed-0 models, 1,000 samples each. Δ is cosine minus linear with a paired bootstrap interval, then at seeds 1, 2. Bold: better by the fixed rule.

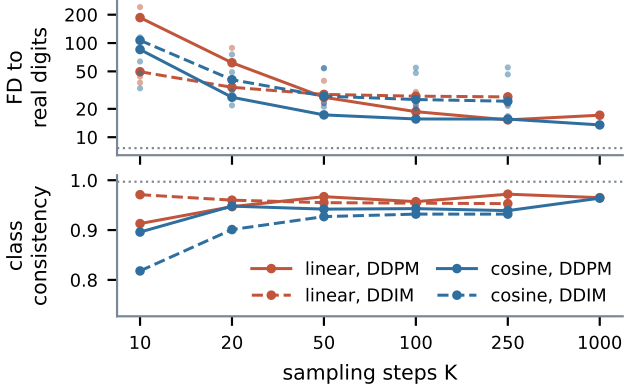


Figure 2: FD (top) and class consistency (bottom) against K at seed 0, with seeds 1 and 2 as small dots; the dotted lines are real test digits.

(open and closed 4s, looped and plain 2s). The linear model draws even strokes much like real digits; the cosine model’s are heavier, and more of its samples are blotted or ambiguous (filled 0s, 8s that read as 6s), hence its lower grid score, though on 1,000 samples the two score alike (0.9650 and 0.9640). The spread agrees: at 1000 steps the cosine samples vary a little more than real digits (1.075 of their spread) and the linear ones a little less (0.959).

With DDPM sampling, the cosine schedule is better at $K = 10$ and 20 and neither is clearly better at $K = 50$, 100 and 250. With DDIM sampling, neither schedule is clearly better at any $K < 1000$. With DDPM the linear model comes within 10% of its own 1000-step FD from $K = 100$, the cosine model at no reduced K , and the cosine model reaches the linear model’s from $K = 100$; with DDIM none of these happens at any reduced K . At the full 1000 steps the FDs are 17.12 (linear) and 13.50 (cosine), a difference of $-3.61 [-7.70, +0.74]$, within evaluation noise.

A follow-up at seed 0, run after the verdict, locates the DDPM gain. At $K = 10$ the second-to-last step jumps to step 0 and leaves noise of standard deviation 0.355 (linear) against 0.185 (cosine), which the last step barely removes. With Ho et al.’s smaller posterior variance $\hat{\beta}$, the FDs at $K = 10$ go from 186.26 to 42.40 (linear) and from 85.35 to 30.64 (cosine), so most of the cosine schedule’s advantage there came from that leftover noise. The cosine schedule thus helps the tutorial’s sampler at few steps mostly through its gentle start near $t = 0$, which shortens that jump, rather than through the gentler noisy end the hypothesis rests on. DDIM adds no noise, and with it neither schedule is reliably better: at $K = 10$ seed 0 favours the linear model (+57.33) and seed 2 the cosine (-4.93). At $K = 10$ the linear model’s DDPM samples also lose variety (spread 0.632, cosine 0.944), and the cosine samples are read as their digit less often at $K = 50$ to 250 with DDPM (0.9430 against 0.9570 at $K = 100$) and at every K with DDIM (0.8180 against 0.9710 at $K = 10$), in line with Figure 1’s heavier strokes.

Training time and image quality. An epoch took 7.87 s (linear) and 7.86 s (cosine), the same within timing

noise, as expected: the schedule changes which noise level a step uses, not the computation. The seed-0 snapshots reach within 10% of their final FD by epoch 150 (linear, 19.7 min) and 100 (cosine, 13.1 min). The final training losses differ (0.0283 linear, 0.0492 cosine at seed 0) mostly because the schedules average over different noise levels: 7 of 20 evenly spaced steps leave under 1% of the signal with the linear schedule and 2 with the cosine, and there the noise is almost free to predict (MSE 0.0007 against 0.0641 elsewhere). At the same noise level with over 1% of the signal left, the cosine model’s error averages 1.00 times the linear model’s (0.83 to 1.40), about the same.

Strengths and limitations. The design isolates the schedule (paired weights, batches, noise and samples), fixes its rule in advance and scales the FD with a real-data floor. Its main limitation is checkpoint noise: each model is its final checkpoint, without weight averaging, so its quality moves between epochs (linear FD 44.72, 71.17 and 28.48 at epochs 50, 100 and 150) and between seeds (cosine, DDPM, $K = 250$: 46.43 at seed 1, 15.58 at seed 0). That spread, not the sample count, leaves most step counts unclear and makes the epoch figures above a single noisy reading. The intervals cover seed 0’s sampling only; the FD uses a small classifier’s features at 1,000 samples, so it is not comparable with FID; and every model is a $T = 1000$ U-Net on 20×20 digits.

Recommendations. Average the weights during training, as Ho et al. do (an exponential moving average, decay 0.9999), and add seeds. Sample in few steps with the posterior variance $\hat{\beta}$ or with learned variances, which Nichol and Dhariwal found keep quality with an order of magnitude fewer steps. To test a smaller T , rescale the linear β range with T , as their released code does; the tutorial’s ends at $\bar{\alpha}_T = 0.364$ over 100 steps.

5. Conclusion

The evidence supports the hypothesis in part: by a rule fixed before the results, the cosine schedule gave better samples at $K = 10$ and 20 with DDPM, and the linear schedule at no reduced step count. Most of that gain comes from the noise the tutorial’s sampler leaves after its last coarse step, which the posterior variance removes; with DDIM neither schedule is reliably better. A stronger claim needs averaged weights and more seeds.

References

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